

CS201 Final

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Abstract—Write only ONE function per question in C++. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures, (iv) you can use any function given in the data structure.

I. QUESTION (ALGORITHM ANALYSIS), 20 POINTS

```

1 int algorithm(int N){
2     if (N == 0)
3         return 0;
4     int sum = 0;
5     for (int i = 0; i < N; i++)
6         for (int j = 0; j < N; j++)
7             for (int k = 0; k < N; k++)
8                 sum++;
9     for (int i = 0; i < 4; i++)
10        for (int j = 0; j < 4; j++)
11            if ((i + j) % 2 == 0)
12                sum += algorithm1(N / 2);
13        else
14            sum++;
15    return sum;
16 }
```

What is the time complexity of algorithm? You **must** construct the recurrence equation and solve it with the master theorem.

II. QUESTION (LINKED LIST), 15 POINTS

Write the method

```
void addAfterEachNode(int X)
```

which adds new node with value X after each node in the singly linked list. You may not use any additional data structure or array.

III. QUESTION (GRAPH), 15 POINTS

Write the method in adjacency list implementation of Graph class

```
Graph* intersection(const Graph& g2,
                   int v)
```

that returns a new graph formed by adding edges which exist both in the original graph and g2 at vertex v. You may assume both graphs are unweighted. You may not use any additional data structure other than the returned graph.

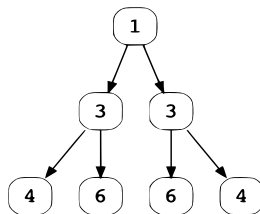
part, the numbers that are equal to X will be in the second part, and the numbers that are larger than X will be in the third part. You may not use any additional data structure or array.

IV. QUESTION (TREE), 15 POINTS

Given a binary tree (not necessarily search tree), implement method

```
bool isMirror(TreeNode* left,
                TreeNode* right)
```

in TreeNode class to check whether an input binary tree is a mirror of itself (symmetric). You may not use any additional data structure or array. Below is an example of a symmetric tree.



V. QUESTION (SORTING), 20 POINTS

Given array A of N numbers write function

```
int maxSortLength(int* A, int N)
```

that determines the maximum length of the sorted sublist in the array. A sublist start from index i and continues with indexes $i + 1, i + 2, \dots, i + k$. Your function should run in $O(N)$ time. You may not use any additional data structure or array.

VI. QUESTION (SORTING), 15 POINTS

Suppose you are given A, an unsorted array of N integers, and a number X. Write $O(N)$ algorithm

```
void threePartitionArray(int* A,
                        int N, int X)
```

to partition the numbers in the array such that, the numbers that are smaller than X will be in the first